NutriFlow

Meet the Team

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Problem Statement

Many people want to eat healthier, save time, and meet their nutrition goals, but meal planning is often fragmented across multiple apps, notes, and grocery lists. This creates additional friction in the form of inconvenience or cost, making healthy habits harder to establish.

Users may track calories in one place, store recipes somewhere else, and build shopping lists manually, making the process inefficient and difficult to maintain.

NutriFlow was designed to address this gap by creating a more unified and accessible meal-planning experience, designed to help users achieve their goals in a sustainable manner.

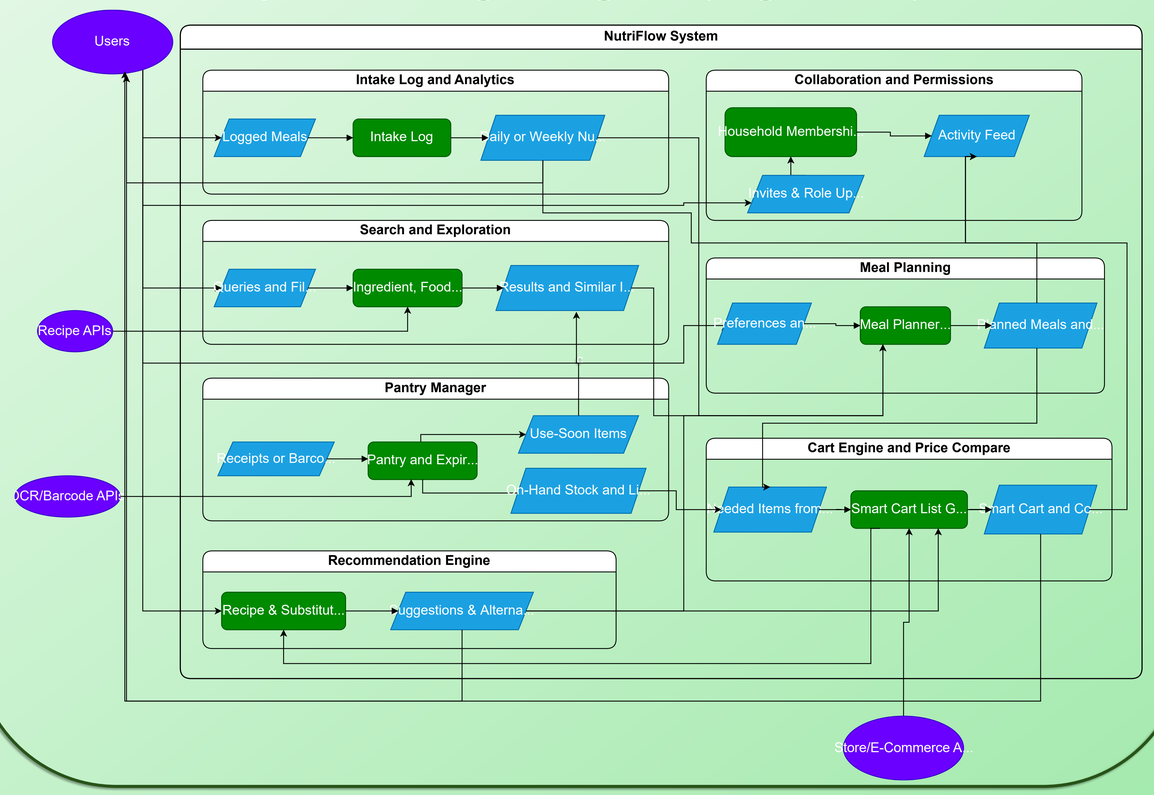
Solution

NutriFlow is a meal-planning and nutrition management platform designed to help users **design meal plans, organize recipes**, **manage pantry inventory**, and **prepare grocery lists** in one system. The project focused on making nutrition planning more practical by connecting these workflows instead of treating them as separate tasks. By combining personalized planning features with a clean, structured user experience, NutriFlow aims to help users make healthier, more cost-effective food decisions and reduce food waste.

Accomplishments

- Designed and developed a full-stack nutrition and meal-planning application
- Created workflows for meal planning, pantry management, and grocery preparation
- Built a user-centered interface focused on clarity and ease of use
- Integrated many technologies across frontend, backend, and data layers to ensure smooth operation across web and mobile platforms
- Applied software engineering principles to create an extensible architecture to handle future expansion

System Design Diagram (High Level)



Technologies Used

Nativewind, gluestack, npm, TS, React Native, Expo, RE, TS, React Query, OPENAPI INITIATIVE, spoonacular, Zod, docker, Prisma, PostgreSQL

Frontend | API/Backend | Database

Challenges

One of the main challenges in developing NutriFlow was designing a system that connects related features without making the user experience feel overwhelming. Meal planning, pantry tracking, recipe organization, and shopping list creation all depend on shared data, so maintaining consistency across the application requires careful system design.

Another challenge was balancing feature scope with usability, ensuring the product remained intuitive while still supporting meaningful functionality.

Finally, ensuring the experience is seamless on both web and mobile platforms introduced additional complexity in terms of what technologies we used and how.

Future Work

Future development for NutriFlow includes expanding personalization features, improving automated meal recommendations, and adding deeper nutritional analysis. We also plan to strengthen household collaboration, enhance pantry-aware planning, and improve grocery integration by supporting additional store chains and more flexible shopping workflows.

Another area of future work is refining the user experience so that planning meals, tracking nutrition, and organizing groceries feel even more seamless and connected. Long term, we hope the project continues to evolve into a comprehensive platform that supports users not only in planning meals, but also in building healthy habits that are sustainable in both a financial and environmental sense.